

Chacha Chen

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I am driven to build better AI that works well with/for people. My research spans applied machine learning, natural language processing, and computational social sciences. Recently, I've been focusing on multimodal LLMs and their applications for expert tasks.

EDUCATION **University of Chicago** 2021 – present
 Ph.D. in Computer Science
 Advisor: [Chenhao Tan](#)

Pennsylvania State University 2019 – 2020
 Master in Informatics
 Advisors: [Ting-Hao \(Kenneth\) Huang](#), [Zhenhui \(Jessie\) Li](#)

Shanghai Jiao Tong University 2015 – 2019
 Bachelor of Science in Computer Science

PROFESSIONAL EXPERIENCE **Microsoft Research**, Research Intern Jun. 2023 –
 Mentors: [Siddharth Suri](#), [Scott Counts](#), [Longqi Yang](#).

Amazon AWS, Applied Scientist Intern Jun. 2022 – Aug. 2022
 Mentors: Sandesh Swamy, Narges Tabari, and [Rashmi Gangadharaiah](#)

IQVIA, Machine Learning Research Intern Jun. 2020 – Aug. 2020
 Mentors: [Cao \(Danica\) Xiao](#), [Fenglong Ma](#) (PSU), and [Jimeng Sun](#) (UIUC).

PUBLICATIONS  [Google scholar](#)

Preprints

1. **Chacha Chen**, Han Liu, Jiamin Yang, Benjamin M. Mervak, Bora Kalaycioglu, Grace Lee, Emre Cakmakli, Matteo Bonatti, Sridhar Pudu, Osman Kahraman, Gul Gizem Pamuk, Aytekin Oto, Aritrick Chatterjee, Chenhao Tan.
Human-AI Decision Making with Domain Experts: A Case Study on AI-Assisted Prostate Cancer MRI Diagnosis
2. Siddharth Suri, Scott Counts, Leijie Wang, **Chacha Chen**, Mengting Wan, Tara Safavi, Jennifer Neville, Chirag Shah, Ryen W White, Reid Andersen, Georg Buscher, Sathish Manivannan, Nagu Rangan, Longqi Yang
The Use of Generative Search Engines for Knowledge Work and Complex Tasks
3. Xiaou Liu, Zhen Lin, Longchao Da, **Chacha Chen**, Shubhendu Trivedi, Hua Wei
MCQA-Eval: Efficient Confidence Evaluation in NLG with Gold-Standard Correctness Labels

Conference Papers

- P.1 **Chacha Chen***, Yuyang Jiang*, Dang Nguyen, Benjamin M. Mervak, Chenhao Tan.
GPT-4V Cannot Generate Radiology Reports Yet. In *Findings of Machine Learning for Health (ML4H 2024)*, *Findings of North American Chapter of the Association for Computational Linguistics (NAACL 2025)*.

- P.2 Nguyen Dang, **Chacha Chen**, He He, and Chenhao Tan. Pragmatic Radiology Report Generation. In *Machine Learning for Health (ML4H 2023)*.
- P.3 **Chacha Chen***, Shi Feng*, Amit Sharma, Chenhao Tan. Machine Explanations and Human Understanding. *Transactions on Machine Learning Research (TMLR 2023)*; Presented at *ACM Conference on Fairness, Accountability, and Transparency (FAccT 2023)*. **Best Paper Award** at the ICML Workshop on Human-Machine Collaboration and Teaming.
- P.4 Vivian Lai, **Chacha Chen**, Alison Smith-Renner, Vera Liao, and Chenhao Tan. Towards a Science of Human-AI Decision Making: A Survey of Empirical Studies. In *ACM Conference on Fairness, Accountability, and Transparency (FAccT 2023)*.
- P.5 Vivian Lai*, Yiming Zhang*, **Chacha Chen**, Vera Liao, and Chenhao Tan. Selective Explanations: Leveraging Human Input to Align Explainable AI. In *ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW 2023)*.
- P.6 Han Liu, Yizhou Tian, **Chacha Chen**, Shi Feng, Yuxin Chen, Chenhao Tan. Learning Human-Compatible Representations for Case-Based Decision Support. In *International Conference on Learning Representations (ICLR 2023)*.
- P.7 Sandesh Swamy, Narges Tabari, **Chacha Chen**, Rashmi Gangadharaiiah. Contextual Dynamic Prompting for Response Generation in Task-oriented Dialog Systems. In *European Chapter of the Association for Computational Linguistics (EACL 2023)*.
- P.8 Chi-Yang Hsu, Yun-Wei Chu, Vincent Chen, Kuan-Chieh Lo, **Chacha Chen**, Ting-Hao Huang, Lun-Wei Ku. Learning to Rank Visual Stories From Human Ranking Data. In *Annual Meeting of the Association for Computational Linguistics (ACL 2022)*.
- P.9 Guanjie Zheng, Chang Liu, Hua Wei, Porter Jenkins, **Chacha Chen**, Tao Wen, Zhenhui Li. Knowledge-based Residual Learning. In *International Joint Conference on Artificial Intelligence (IJCAI 2021)*.
- P.10 **Chacha Chen**, Junjie Liang, Fenglong Ma, Lucas Glass, Jimeng Sun, Cao Xiao. UNITE: Uncertainty-based Health Risk Prediction Leveraging Multi-sourced Data. In *The Web Conference (WWW 2021)*.
- P.11 Guanjie Zheng, Chang Liu, Hua Wei, **Chacha Chen**, Zhenhui Li. Rebuilding City-Wide Traffic from Multi-Source Data. In *IEEE International Conference on Data Engineering (ICDE 2021)*.
- P.12 Hua Wei, **Chacha Chen**, Chang Liu, Guanjie Zheng and Zhenhui Li, ImIn-GAIL: Learning to Simulate with Imitation-Interpolation on Sparse Trajectory Data. In *European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD 2020)*.
 **Best Applied Data Science Paper Award**
- P.13 **Chacha Chen**, Hua Wei, Nan Xu, Guanjie Zheng, Ming Yang, Yuanhao Xiong, Kai Xu and Zhenhui Li, Toward A Thousand Lights: Decentralized Deep Reinforcement Learning for Large-Scale Traffic Signal Control. In *Proceedings of the Thirty-Fourth AAAI Conference on Artificial Intelligence (AAAI 2020)*.
- P.14 Hua Wei, Nan Xu, Huichu Zhang, Guanjie Zheng, Xinshi Zang, **Chacha Chen**, Weinan Zhang, Yanmin Zhu, Kai Xu and Zhenhui Li, CoLight: Learning Network-level Cooperation for Traffic Signal Control. In *Proceedings of the 2019 ACM on Conference on Information and Knowledge Management (CIKM 2019)*.
- P.15 Hua Wei, **Chacha Chen**, Guanjie Zheng, Kan Wu, Vikash V. Gayah, Kai Xu and Zhenhui Li, PressLight: Learning Max Pressure Control to Coordinate Traffic Signals in Arterial Network. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD 2019)*.

Workshop Papers

W.1 **Chacha Chen***, Tongan Cai*, Ting-Hao K. Huang, Frank E. Ritter, What Makes A Good Reference Manager? A Quantitative Analysis of Bibliography Management Applications, Asian CHI Symposium 2021 (**AsianCHI2021**).

 **Best Paper Award**

W.2 **Chacha Chen**, Guanjie Zheng, Hua Wei, Zhenhui Li, Physics-informed Generative Adversarial Networks. In Proceedings of the NeurIPS Workshop on Interpretable Inductive Biases and Physically Structured Learning (**IIBPSL, NeurIPS 2020**).

AWARDS	6th place, <i>PICAI grand challenge</i>	Aug. 2023
	Crerar Fellowship, <i>The University of Chicago</i>	Feb. 2021
	AAAI-20 VISA Research Scholarship	Feb. 2020
	AAAI-20 Student Travel Grant	Feb. 2020
	Top 25 in the finalist, <i>KDD Cup 2020 LDR Competition</i>	Aug. 2020
	Distinguished Bachelor Thesis, <i>Shanghai Jiao Tong University</i>	Jun. 2019
	Academic Excellence Scholarship, <i>Shanghai Jiao Tong University</i>	Sep. 2016
COMMUNITY SERVICE	Organizing committee of FAccT conference 2024, 2025.	
TALKS	<i>Machine Explanations and Human Understanding</i>	
	Human-Centric AI group at NEC Labs Europe in Heidelberg, Germany, 2022.	
	<i>Interpretable Machine Learning</i>	
	Data Mining and Machine Learning seminar at the Pennsylvania State University, 2020	
TEACHING EXPERIENCE	The University of Chicago	
	Teaching Assistant	Spring 2024
	CMSC 25300: Mathematical Foundations of Machine Learning	
	Prof. Haifeng Xia, Prof. Hannah Morgan	
	Pennsylvania State University	
	Teaching Assistant	Fall 2020
	IST 261 (Java Course): Application Development Design Studio I	
	Prof. Kenneth Huang	